

ELDRED ROCK, AK, USA

WMO: 997789

Lat: 58.9705N

Lon: 135.2200W

Elev: 62

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.3	(c) 14.5	(d) 0.1	(e) 5.5	(f) 13.8	(g) 2.6	(h) 6.3	(i) 17.5	(j) 60.4	(k) 16.5	(l) 57.4	(m) 19.6	(n) 40.7	(o) 330	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	69.1	58.2	66.2	57.5	63.9	56.8	60.1	65.9	58.9	64.4	57.8	62.7	6.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.3	62.4	56.2	67.4	61.4	55.1	64.8	60.3	26.4	66.1	25.6	64.5	24.9	63.0	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 52.6	(b) 47.5	(c) 41.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	31.5	31.8	34.9	41.4	50.3	56.0	58.0	57.4	51.8	44.2	35.5	31.7		
	DBStd	11.49	8.82	6.98	5.53	3.98	5.21	4.10	3.64	3.36	3.27	3.61	6.06	6.78		
	HDD50	3068	575	509	467	257	60	2	0	0	17	179	434	568		
	HDD65	7751	1040	929	932	707	456	272	220	237	397	644	884	1033		
	CDD50	801	0	0	0	1	70	182	248	229	70	1	0	0		
	CDD65	5	0	0	0	0	0	2	3	0	0	0	0	0		
	CDH74	20	0	0	0	0	0	6	14	0	0	0	0	0		
	CDH80	2	0	0	0	0	0	1	1	0	0	0	0	0		
	WSAvg	18.2	24.2	23.5	19.7	14.6	13.3	14.1	13.6	13.1	17.3	20.8	22.8	22.1		
	PrecAvg	71.4	7.4	5.8	5.2	3.6	3.0	2.4	3.3	4.6	8.5	10.2	8.7	8.7		
(15) Precipitation	PrecMax	95.4	13.2	9.3	11.7	6.7	6.1	4.4	6.1	8.6	12.0	16.2	16.1	17.1		
	PrecMin	58.0	3.3	2.0	0.8	1.0	0.7	0.9	1.0	1.7	3.0	4.7	2.8	4.5		
	PrecStd	8.9	2.4	2.1	2.2	1.4	1.4	0.8	1.3	1.7	1.9	2.6	2.8	3.4		
	PrecStd	8.9	2.4	2.1	2.2	1.4	1.4	0.8	1.3	1.7	1.9	2.6	2.8	3.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	43.4	49.7	56.1	68.5	73.5	75.3	72.3	63.5	52.9	45.8	42.5		
		MCWB	43.1	39.3	42.4	45.2	54.1	57.8	58.4	57.7	52.7	48.5	42.4	40.4		
	2%	DB	44.1	42.1	45.1	52.2	64.2	68.6	68.3	67.5	60.6	51.0	44.3	41.7		
		MCWB	41.7	39.0	40.7	43.6	54.2	57.2	59.8	58.0	53.0	47.2	41.5	39.5		
	5%	DB	42.3	40.9	42.7	49.4	60.9	64.9	65.4	65.0	57.7	49.4	43.2	40.9		
		MCWB	40.4	38.4	39.2	42.5	52.6	56.2	58.3	58.1	52.9	45.9	40.9	38.8		
	10%	DB	40.6	39.5	41.2	46.8	58.1	62.2	63.2	62.6	56.0	48.5	42.2	39.9		
		MCWB	38.9	37.1	38.3	41.3	50.8	55.2	57.6	57.0	52.3	45.2	40.0	37.9		
	0.4%	WB	44.2	41.3	44.1	46.4	57.5	61.1	62.8	60.5	56.5	50.6	43.4	41.4		
		MCDB	46.2	42.1	48.0	53.2	64.7	68.4	68.8	65.4	59.3	51.9	44.8	41.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.3	40.2	41.8	44.4	55.4	58.8	60.6	59.5	55.1	48.4	42.1	40.1		
		MCDB	43.6	41.3	44.4	50.3	62.4	66.7	66.6	64.6	57.8	50.2	43.5	41.0		
	5%	WB	40.7	39.1	40.1	43.2	53.6	57.3	59.2	58.6	54.0	47.2	41.2	39.2		
		MCDB	41.8	40.4	42.3	48.1	60.2	64.3	64.4	63.3	56.6	49.1	42.6	40.4		
	10%	WB	39.2	37.5	38.7	42.1	51.6	55.8	58.0	57.5	52.9	46.1	40.2	38.2		
		MCDB	40.4	39.1	40.8	46.1	57.3	61.4	62.5	61.7	55.4	48.1	41.7	39.5		
	5% DB	MDBR	5.6	5.6	6.1	7.8	9.0	8.2	7.2	7.0	5.7	4.8	5.0	5.1		
		MCDBR	7.0	6.2	7.9	12.0	14.6	14.8	13.0	12.4	8.8	6.0	5.0	5.7		
		MCWBR	5.2	4.7	5.3	6.6	7.4	7.5	7.2	6.7	4.7	4.1	4.0	4.7		
		MCWBR	6.8	5.7	7.6	10.1	12.8	13.0	11.8	9.9	7.6	6.1	5.0	5.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.2	4.7	5.5	6.0	7.4	7.2	6.9	6.1	4.6	4.7	4.1	5.1		
		MCWBR	5.2	4.7	5.5	6.0	7.4	7.2	6.9	6.1	4.6	4.7	4.1	5.1		
	Clear-Sky Solar Irradiance	taub	0.285	0.304	0.325	0.372	0.387	0.413	0.417	0.383	0.353	0.315	0.273	0.253		
		taud	2.021	2.096	2.076	1.985	2.055	2.089	2.153	2.280	2.312	2.237	2.054	1.984		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	153	219	252	259	265	259	255	256	245	215	156	122		
		Edh at Noon	17	26	36	47	47	46	43	35	29	23	16	12		
(44) All-Sky Solar Radiation	RadStd	RadAvg	122	357	763	1271	1661	1616	1422	1120	716	356	161	80		
		RadStd	13	59	90	135	162	110	132	127	93	49	23	8		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	+3.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page